



Communication

Detection of intermolecular homonuclear dipolar coupling in organic rich shale by transverse relaxation exchange

Kathryn E. Washburn^{a,*}, Yuesheng Cheng^b^a Ingrain, Inc., 3733 Westheimer Road, Houston, TX 77027, United States^b Halliburton, 3000 North Sam Houston Pwky E, Houston, TX 77032, United States

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ABSTRACT

The mechanism behind surface relaxivity within organic porosity in shales has been an unanswered question. Here, we present results that confirm the existence of intermolecular homonuclear dipolar coupling between solid and liquid phases in sedimentary organic matter. Transverse magnetization exchange measurements were performed on an organic-rich shale saturated with liquid hydrocarbon. Liquid and solid constituents were identified through both sample resaturation and through their T_1/T_2 ratios. Extensive cross peaks are observed in the T_2 - T_2 exchange spectra between the solid and liquid constituents, indicating an exchange of magnetization between the two phases. This result cannot arise from physical molecular diffusion, and the dissolution energies are too high for chemical exchange, such that the magnetization exchange must arise from intermolecular homonuclear dipolar coupling. These results both confirm a possible source of surface relaxivity in organic matter and emphasize caution in the use of standard porous media interpretations of relaxation results in shales because of coupling between different magnetization environments.

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1. Introduction

In the past decade, numerous investigations of unconventional petroleum resources [1–13] by nuclear magnetic resonance (NMR) have been conducted. The low permeability of shales makes traditional core analysis characterization difficult and unreliable [14], such that a non-invasive method such as NMR is appealing. Despite its extensive use, many unanswered questions still exist regarding the interpretation of the NMR results in shale and mudrocks.

One of these questions is the mechanism of surface relaxivity in organic matter [15]. Surface relaxivity in conventional petroleum reservoirs [16–23] has been well established as arising from the presence of paramagnetic impurities on the pore surface. When a fluid molecule encounters a paramagnetic ion, the interaction between the electron and nuclear spins causes the rapid relaxation of the nuclear spin. Shales may contain a significant quantity of paramagnetic impurities; however, in organic matter itself, the amount of paramagnetic impurities is extremely low [24,25]. Given their low concentrations, it does not seem plausible that

the dominant mechanism of surface relaxivity in organic porosity is due to fluid interaction with paramagnetic impurities.

A significant difference between unconventional shale resources and conventional petroleum resources is the presence and potentially high concentrations of organic hydrogen in solids. For conventional resources, the pore matrix consists predominantly of either sandstone or carbonate, neither of which contain significant quantities of hydrogen. Therefore, any hydrogen in the sample is assumed to be in fluid-filled porosity. However, organic matter is rich in hydrogen and, in many cases, it is the organic matter that contains most of the porosity in the sample [26]. Several researchers have hypothesized that the surface relaxivity arises from the homonuclear dipolar coupling [2,9,15] between the hydrogen in the organic matter and the saturating fluid. Korringa, SeEVERS, and Torrey presented the theory of longitudinal surface relaxation for the interaction between fluid molecules and paramagnetic ions [16]. This was later adapted for transverse relaxation by Kleinberg et al. [17]. This has been adapted to surface relaxation due to homonuclear dipolar coupling [27]. If two spins are undergoing dipolar coupling, this will cause an enhancement in their relaxation rate. For a fluid molecule adsorbed on the pore surface, the two spins will be in the rigid lattice regime and this enhancement will be inversely proportional to the strength of their coupling [28]:

* Corresponding author.

E-mail address: Kathryn.washburn@nofima.no (K.E. Washburn).

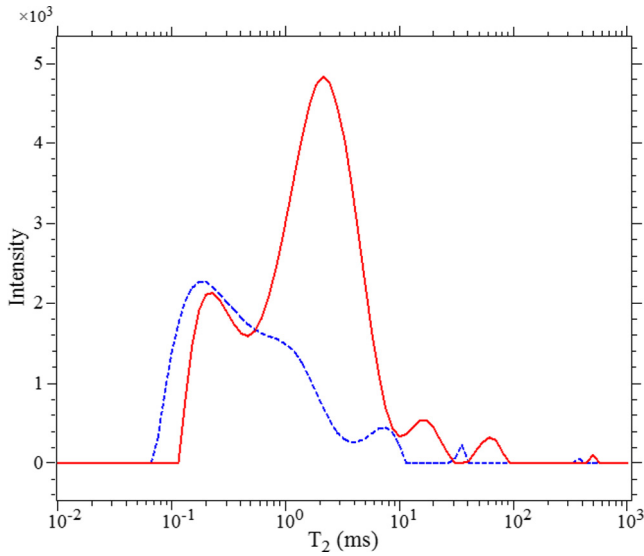


Fig. 1. T_2 distribution for the (a) as-received state (blue dashed) and (b) resaturated state (solid red) (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

$$\frac{1}{T_{2D}} = \|D_{ij}\| = \left\| -\frac{\mu_0 \gamma^2 \hbar}{4\pi r_{ij}^3} \left(\frac{3 \cos^2 \theta_{ij} - 1}{2} \right) \right\| \quad (1)$$

where T_{2D} is the relaxation time of two hydrogen atoms undergoing dipolar coupling, D_{ij} is the dipolar coupling constant, the brackets denote the absolute value, μ_0 is the permeability of a vacuum, γ is the gyromagnetic ratio, $\hbar$ is Planck's constant, r_{ij}^3 is the distance between the coupled spins, and θ_{ij} is the angle they make with the applied magnetic field. While Korringa, Seever, and Torrey theory contains a complete description of how surface correlation time affected the measured relaxation time, Kleinberg et al. [17] later found that for transverse relaxation due to paramagnetic ions, the correlation term could be neglected because the relaxation time between the nuclear spin of the fluid molecule and the electron spin of the paramagnetic ion was significantly more rapid than the surface residence time. However, the relaxation time from the dipolar coupling cannot be assumed to be much greater than the surface

residency of the fluid molecule, such that the correlation time must be taken into consideration. Therefore, the transverse relaxation rate for a single pore in the fast diffusion limit is described as:

$$\frac{1}{T_2} = \frac{1}{T_{2B}} + \left(\frac{Sh}{V} \right) \left(\frac{n_H}{T_{2D} + \tau_c} \right) \quad (2)$$

where T_2 is the measured relaxation time, T_{2B} is the relaxation time of the bulk fluid, S is the surface area, V is the pore volume, h is the layer thickness, n_H is the relative fraction of hydrogen sites on the surface of the pore surface, and τ_c is the correlation time of how long a fluid molecule remains at the surface before exchanging with the bulk. Eq. (2) can be simplified by rewriting it to include a surface relaxivity term:

$$\rho_D = \frac{h \times n_H}{T_{2D} + \tau_c} \quad (3)$$

where ρ_D is surface relaxivity due to dipolar coupling. Then, assuming the system contains a low viscosity fluid, the bulk term can be neglected and the system can be described in a form similar to that of transverse relaxation due to paramagnetic ions:

$$\frac{1}{T_2} = \rho_D \frac{S}{V} \quad (4)$$

Although there were many indications that homonuclear dipolar coupling was the source of surface relaxivity in organic matter, there was no direct evidence. For example, if the surface relaxivity is caused by homonuclear dipolar coupling, because of the presence of τ_c in ρ_D , the measured transverse relaxation time will become longer with increased temperature due to shorter correlation times. Longer transverse relaxation times with increased temperature in organic rich shales were observed [29], but other possible causes for the longer relaxation times could not be ruled out, such as that the change in relaxation time was caused by a decrease in oil viscosity with temperature. Here, the molecular rotational correlation time decreases with temperature, leading to weaker dipolar interactions within the oil molecules themselves and a longer relaxation time would be observed regardless of whether the source of surface relaxivity in the porous sample were due to interactions with paramagnetic impurities or homonuclear dipolar coupling.

Recent work by Singer et al. [30] has shown the existence of intramolecular homonuclear dipolar coupling in fluid-filled organic matter. The method used was an adaptation of the tech-

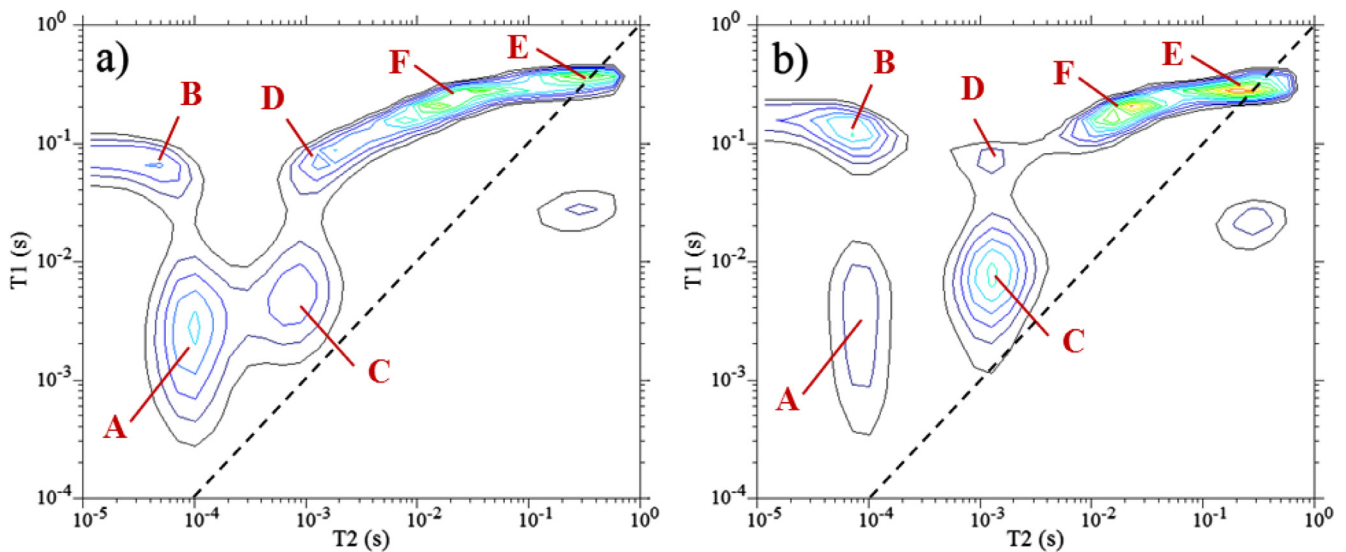


Fig. 2. T_1 - T_2 correlation for the (a) as received state sample and (b) resaturated state sample. Dashed line indicates parity.

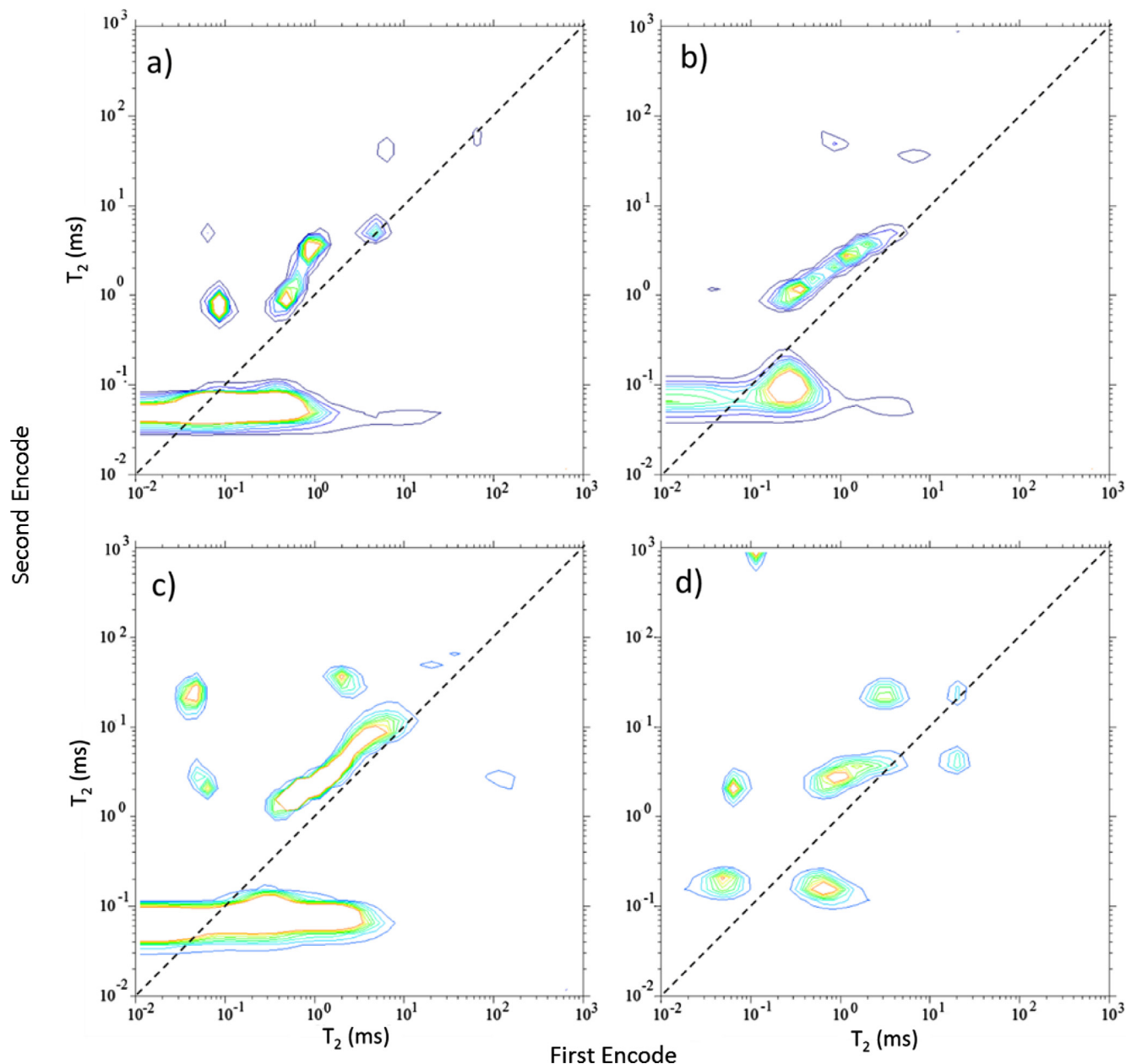


Fig. 3. T_2 - T_2 exchange plots for the as-received state sample at (a) 100 μ s (b) 1 ms (c) 5 ms (d) 10 ms mixing times.

nique by Straley [31] to confirm the existence of homonuclear dipolar coupling in carbonates. To avoid an influence from inorganic pores, the organic matter was isolated from the shale matrix. The kerogen was then saturated with both normal distilled water and a mix of normal and deuterated water. If the relaxation was due to paramagnetic impurities, the relaxation time of the two systems should be identical. However, if the relaxation was due to homonuclear dipolar coupling, there should be an increase in the relaxation time of the partially deuterated system, which was observed. This method confirms the existence of intramolecular homonuclear dipolar coupling; however, it does not confirm the presence of intermolecular homonuclear dipolar coupling.

This manuscript presents evidence to confirm the existence of intermolecular homonuclear dipolar coupling between fluid molecules and organic matter. Transverse relaxation exchange, also known as T_2 - T_2 exchange [32,33], is a pulse sequence that measures the movement of magnetization between different environ-

ments. The pulse sequence begins with a partial Carr Purcell Meiboom Gill (CPMG) [34,35] sequence to encode for T_2 , and then the magnetization is stored along the z-axis for a period of time.

During this mixing time, the system does not experience transverse relaxation, only longitudinal relaxation, and the fluid molecules are allowed to diffuse between environments. After the mixing time, the magnetization is returned to the transverse plane, and a full CPMG measurement is acquired. The experiment is repeated, increasing the number of 180° pulses in the initial partial CPMG before the storage time. The experiment is then analyzed by a two-dimensional inverse Laplace transform [36,37]. If magnetization remains in its original environment, then its signal occurs on the diagonal line of the 2D T_2 - T_2 map, indicating that the T_2 environment for the magnetization was the same at the end of the experiment as in the beginning. If there is no exchange between T_2 environments or if the mixing time is much shorter than the rate or rates of exchange in the system, all signal in the

T_2 - T_2 map will be located along the diagonal [38]. The formation of cross peaks indicates that magnetization has moved from one T_2 environment to another T_2 environment during the mixing time.

Although the method is generally used to investigate the diffusion or flow of spin-bearing fluid molecules between different pore environments, it can be used to observe the movement of magnetization between any environments with different transverse relaxation times. To this end, we sought to find evidence of intermolecular homonuclear dipolar coupling by performing a T_2 exchange measurement on an organic-rich shale saturated with liquid hydrocarbon. Besides molecular diffusion, another mechanism that can lead to the exchange of transverse relaxation is spin diffusion. When two spins are undergoing dipolar coupling, there is a possibility that they may swap energy states, a process called spin diffusion. The T_2 - T_2 exchange pulse sequence can as readily measure the change of the relaxation environment due to spin dif-

fusion as it can measure an exchange resulting from molecular diffusion. The presence of cross peaks between solids and liquids in the spectra provides a confirmation that the constituents are exchanging magnetization. While there is a third possible mechanism for an exchange of magnetization between T_2 environments, chemical exchange [33], the dissolution energies make chemical exchange between solid organic matter and liquid hydrocarbon extremely unlikely [39]. Therefore, we are confident that cross peaks between signal associated with organic solids and liquid hydrocarbon indicate the two constituents are experiencing intermolecular homonuclear dipolar coupling.

2. Experimental

Measurements were performed on a core plug from an organic-rich, liquid hydrocarbon-bearing reservoir. The sample was taken

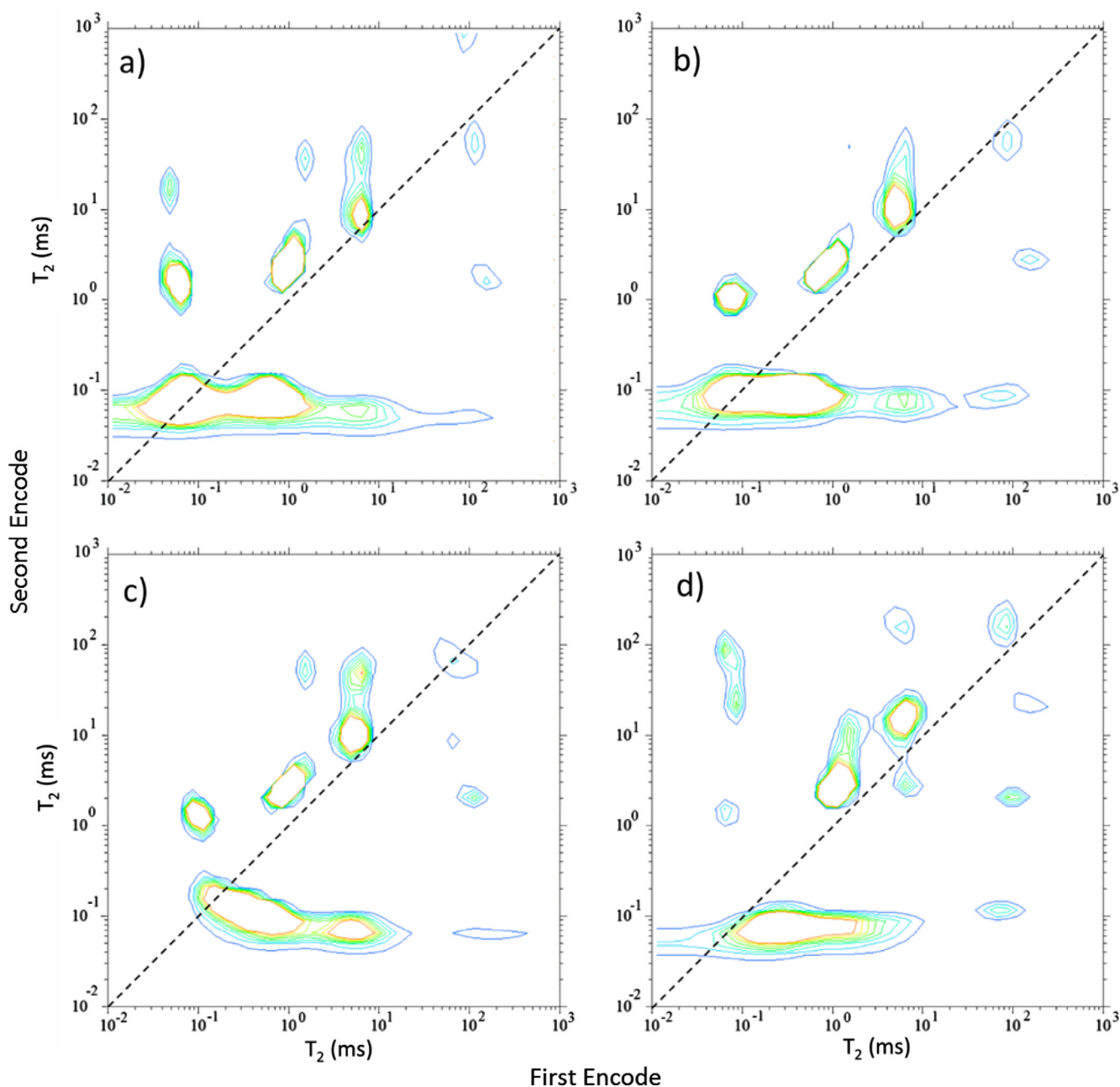


Fig. 4. T_2 - T_2 exchange plots for the resaturated state sample at (a) 100 μ s (b) 1 ms (c) 5 ms (d) 10 ms mixing times.

from a zone that had, on average, 9.8% organic matter by weight, and 2D scanning electron microscope (SEM) imaging confirmed the majority of porosity ($\sim 90\%$) to be located in the organic matter. Focused Ion Beam scanning electron microscopy (FIB-SEM) found average pore sizes ranged from 25 nm to 1 μm and approximately 95% of the pore space was connected with a horizontal permeability of 12.3 μD and a vertical permeability of 0.4 μD . Programmed pyrolysis measurements [40] of a sample end trim confirmed that the sample contained large quantities of both forms of organic matter: kerogen, solid organic matter, and bitumen, semi-solid organic matter. The region from which the sample was taken contained almost no clay ($<5\%$), as confirmed by X-ray diffraction (XRD) and Fourier transform infrared spectroscopy (FTIR). The core plug was measured in both the as-received state and the resaturated state. In the as-received state, no purposeful changes have been made to the state of the core, for example – to clean it, but it is expected that there may be unavoidable changes to the core as compared to its condition in the reservoir, such as evaporation of liquids or empty pore space as a result of gas loss during the core retrieval. Resaturation was performed by placing the sample under a vacuum and then saturating using heptane at a pressure of 2000 psig for a week.

NMR measurements were performed on a GeoSpec (Oxford Instruments, Oxford UK) using GIT System software (Green Imaging Technologies, New Brunswick, Canada). T_2 was measured with the standard CPMG method and 20,000 echoes per measurement were acquired. For the T_2 - T_2 exchange experiments, mixing times of 100 μs , 1 ms, 5 ms, 10 ms, 20 ms, 50 ms, and 100 ms were used. The initial partial CPMG in the T_2 - T_2 exchange pulse sequence was encoded from 0.1 to 250 ms with 32 steps spaced equally on a logarithmic scale. At each step, 2500 data points were acquired. The inversion-recovery method was used for T_1 - T_2 measurements. A total of 32 recovery times were used, ranging from 0.04 to 2277 ms on an equally spaced logarithmic scale. For each recovery time, 20,000 data points were acquired. A 2 s wait time and 0.1 ms echo time were used for all NMR measurements.

Experimental data were analyzed with Prospa© software (Magritek Ltd., Wellington New Zealand) using a Butler-Reed-Dawson inversion for the one-dimensional measurements [41] and a maximal entropy inversion for the two-dimensional inverse Laplace transform [37]. Although previous work has shown that analyses of organic shale relaxation data are best performed with an inversion method that can solve for the Gaussian relaxation of solids and the exponential relaxation of liquids [42], to date a two-dimensional method has not been developed. Despite the appearance of peaks at artificially low relaxation times for solid constituents using a standard two-dimensional inverse Laplace transform, the results are stable and repeatable. Furthermore, all other inversions in the literature are performed with a standard two-dimensional inverse Laplace transform, and therefore this approach best enables a comparison of our results to those from previous researchers.

3. Results and discussion

In previous work to observe the magnetization exchange in organic-rich samples [15,43], it could not be conclusively stated that the exchange was occurring between liquids and solids, as opposed to solids and semi-solids, such as between kerogen and bitumen. To confirm the identity of the solid and liquid constituents in the sample, we compared both the T_2 distributions (Fig. 1) and the T_1 - T_2 correlation plots (Fig. 2) for the sample in the as-received state and the resaturated state.

A comparison of the T_2 distributions for the as-received and resaturated states shows a large increase in the signal located at

~ 2 ms after resaturation, which suggests that this region is predominantly associated with hydrocarbons in organic pores. This result agrees with previous work, as other researchers have observed that this region of the T_2 distribution is associated with hydrocarbon in organic porosity at 2 MHz [44,45]. Two weak peaks exist at relaxation times above 10 ms, a region of the T_2 distribution attributed to hydrocarbon in inorganic porosity. Both peaks shift to slightly to longer relaxation times after resaturation. We interpret this shift to be due to air occupying a portion of the pore space in the as-received sample. The short peak at $\sim 100 \mu\text{s}$, a region of the T_2 spectrum typically associated with organic matter, decreases slightly in intensity and there is a slight shift to a longer relaxation time. This may be due to an averaging of the relaxation time if magnetization exchange occurs between liquids in solids, or it may be simply an artifact of the inversion.

We see in the T_1 - T_2 correlation measurements that there exist two different constituents at the short T_2 relaxation time. Peak A has a T_1/T_2 ratio of approximately 20–30, which is on the same order as previously attributed to bitumen [45]. Surprisingly, a peak also exists at a T_1/T_2 ratio of approximately 1000 (Peak B), which is a ratio associated with solid kerogen at 2 MHz [46]. It was not expected that the kerogen would be detected using a 100 μs echo spacing. However, if there is significant magnetization exchange occurring, it may be that the kerogen T_2 value is being averaged with that of the liquid hydrocarbons, enabling it to be detected. The intensity of the kerogen peak increases slightly in the resaturated sample, which supports this theory. The liquid hydrocarbon peak at $T_2 \sim 2$ ms (Peak C) has a T_1/T_2 ratio of approximately 4, which is also in line with ratios found in the literature. A smaller peak appears above the main oil peak with a longer T_1 time (Peak D). This result may arise from the averaging of magnetization with solid sample constituents during the measurement. As with the kerogen, this peak increases in intensity in the resaturated state.

Two peaks were observed in the region of the T_2 spectra typically attributed to hydrocarbon present in inorganic matter. The peak with the longer T_2 value (Peak E) has a T_1/T_2 ratio close to unity, which is in line with the literature value for hydrocarbon in inorganic porosity. The T_1/T_2 ratio of the oil peak at 20 ms (Peak F) was approximately 10; this result was higher than expected because literature values generally put the ratio of hydrocarbons in inorganic porosity in the range of 1–2 [44]. We believe that, despite the longer relaxation time, this peak arises from hydrocarbon associated with organic porosity. There appears to be a weak peak at nonphysical time ($T_2 > T_1$) in both the as-received and resaturated spectra. We are uncertain as to its origin. It may arise from magnetization exchange between constituents with different T_1/T_2 ratios in the sample with during the course of measurement [47] or perhaps from non-random noise (e.g. noise from electronic circuitry) picked up by the probe.

Based on the T_2 and T_1 - T_2 correlation measurements, we are confident in the identification of the different T_2 peaks as associated with organic solids, hydrocarbons in organic pores, and hydrocarbons in inorganic pores. We then apply this identification to the interpretation of the transverse relaxation exchange plots. Fig. 3 shows the T_2 exchange plots for the as-received state, and Fig. 4 shows the resaturated state. The intensity ranges were kept the same for both samples, such that some of the weaker peaks for the as-received state plots may not be visible. Even at the shortest exchange time, extensive exchange peaks are observed between solid ($T_2 \sim 0.1$ ms) and liquid ($T_2 \sim 2$ ms) constituents in both the as-received and the resaturated sample. This result is in line with previous research, which showed that magnetization exchange occurred as rapidly as the measurement could be made [15,43]. This exchange of magnetization between the solid and liquid constituents confirms the presence of homonuclear dipolar coupling between them, as the exchange could not be the result of molecu-

lar diffusion of spin-bearing fluid molecules. Although spin diffusion rates are slow, the exchange depends on the surface area between the liquids and solids [48]. In shales, the specific surface area is frequently very large [49], although no specific surface area measurement is available for this particular sample. We observe that exchange rapidly occurs between the solids ($T_2 \sim 0.1$ ms) and the peak located at approximately $T_2 \sim 20$ ms, which supports the theory that, despite the long T_2 time, the peak is associated with hydrocarbons in organic porosity.

At the longer exchange times, the as-received and resaturated states develop cross peaks that indicate an exchange of magnetization between the inorganic ($T_2 \sim 100$ ms) and organic porosity ($T_2 \sim 2$ ms and $T_2 \sim 20$ ms) and the two organic porosity environments ($T_2 \sim 2$ ms and $T_2 \sim 20$ ms). This exchange is the result of molecular diffusion of the spin-bearing molecules between pores and suggests strong diffusional coupling in the systems. There also begins to develop cross peaks at longer exchange times between the inorganic porosity ($T_2 \sim 100$ ms) and the organic solids ($T_2 \sim 0.1$ ms). As there is not a direct mechanism for magnetization to exchange occur between organic solids to inorganic porosity, this suggests that multiple exchanges of magnetization are occurring on the time scale of the NMR measurement. These results underscore that the interpretation of NMR relaxation data from organic-rich shales must be treated differently than the standard interpretation for conventional sandstone and carbonate reservoirs. If diffusional coupling is present, the transverse relaxation distribution will not be reflective of the relative pore sizes in the sample. Although the inverted data for the 20 ms, 50 ms, and 100 ms exchange times showed exchange peaks, the signal-to-noise ratio (SNR) was too low (<100 SNR) for the inversion to be considered reliable, and significant T_1 weighting was evident.

Significant asymmetry was observed in some of the exchange spectra, which has been associated with multisite exchange [50]. Given the rapid exchange that occurs between the constituents, it is highly likely that magnetization is exchanging environments multiple times during the course of the measurement. This also suggests that exchange is occurring not only during the mixing time, but during the CPMG encode periods as well. This has been observed to lead to shifts in the diagonal peaks away from the diagonal and asymmetry in the cross peaks [51]. The asymmetry may also be a result of the large T_1/T_2 ratio of the solid sample constituents. Because of the complexity of the exchange in the spectra, no quantitation of exchange rates was attempted here. Additional work involving simulation would be necessary to fully understand the effect of exchange between constituents with low and high T_1/T_2 ratios on the resulting T_2 - T_2 exchange plots.

4. Conclusions

This manuscript presents the first direct evidence for intermolecular homonuclear dipolar coupling in organic porosity. Sample constituents were identified by hydrocarbon resaturation and T_1 - T_2 correlation. Transverse relaxation exchange experiments show clear movement of magnetization between the solid and liquid sample constituents. This, along with recent research on the presence of intramolecular dipolar coupling, helps to understand the source of surface relaxivity in fluid-filled organic matter. In addition to the presence of intermolecular homonuclear dipolar coupling, the transverse relaxation results show diffusional coupling of fluid molecules between organic and inorganic porosity in the sample.

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